



Self-Regulated Learning, Multimodal Data, and Analysis Grid: Where Are We Now and Where Are We Going?

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Abstract

This special issue examines advances in the measurement and support of self-regulated learning (SRL), emphasizing the integration of multimodal data and artificial intelligence (AI) in educational contexts. SRL is a goal-directed process in which learners plan, monitor, and control their learning, influenced by the interplay of cognition, affect, metacognition, and motivation (CAMP). Traditional methods in educational psychology, such as self-reports and interviews, often fall short of capturing the dynamic and recursive nature of SRL. Recent research employs multimodal data and process-oriented approaches to better understand the complex interactions among CAMP processes. The self-regulated learning, multimodal data, and analysis grid (SMA) is used as a framework for mapping and analyzing these processes across diverse data streams. The special issue includes three review papers and two empirical studies that illustrate the benefits and challenges of integrating multiple data sources and analytical techniques, while emphasizing the need for reliable and valid measures to enable personalized support for SRL. Collectively, the studies provide a multidisciplinary perspective on the current state and future directions of SRL research, advocating for innovative, theory-driven approaches that leverage existing technological capabilities to empower agentic learners in digital environments.

Keywords Affect · Artificial intelligence · Cognition · Metacognition · Motivation · Self-regulated learning

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Introduction

Self-regulated learning (SRL) is a goal-oriented process in which learners plan, monitor, and control their learning to achieve their learning goals (Winne & Hadwin, 1998; Zimmerman, 2000). In addition to cognition and metacognition, learners' motivational and affective conditions affect whether and how learners engage in SRL. Even though conventional methods in educational psychology (such as self-reports and interviews) provide important insights on learners' dispositions and situational interpretations of the task and self, these measures alone do not capture the temporal and recursive nature of SRL (Azevedo & Gašević, 2019; Järvelä & Bannert, 2021; Molenaar et al., 2023). Thus, researchers have increasingly adopted process-oriented approaches to study the covert and intertwined nature of learners' cognition, affect, metacognition, and motivation (CAMM).

The field of studying SRL is currently challenged to advance our understanding of how multimodal data can unobtrusively capture the temporality, sequence, and order of learners' CAMM processes. Luckily, the intersection of SRL theory with learning analytics (Roll & Winne, 2015), and artificial intelligence (AI; Järvelä et al., 2023) are promising. The interdisciplinary efforts among researchers in educational psychology, LA, and AI have leveraged SRL-research as a theoretical foundation, focusing on the measurement of SRL through LA and AI (Azevedo & Gašević, 2019). To organize where studies are now by measuring the different modalities of SRL, we have introduced the self-regulated learning, multimodal data, and analysis grid (SMA grid, Molenaar et al., 2023).

The use of (un)obtrusive data streams and advanced statistical and machine learning -based analysis techniques have improved the measurement of SRL with the increasing interest towards integrated approaches, i.e. using multiple data streams to measure multiple CAMM processes within a single study. At the same time, the SMA grid showed how the temporal nature of affective and motivational processes has not gained that much attention when measuring SRL through LA and AI in comparison to the cognitive and metacognitive processes. Thus, further advancements in measuring the combination of CAMM processes and data streams are needed as reliable and valid measurement forms the basis for providing personalized support for these processes.

Although capturing learning trace data in fully digital learning environments often makes providing support more feasible (Bernacki et al., 2025), the SMA grid and process-oriented SRL measurements can also be applied in other environments that include some digital components to offer support (Edwards et al., 2025). While research has investigated the support of learners' SRL through various advanced learning technologies (e.g., intelligent tutoring systems, Azevedo et al., 2022; rule-based AI scaffolds, Lim et al., 2024), recent advancements in generative AI (e.g., large language models) offer new avenues for SRL support through human-AI interactions (Li et al., 2025). However, empirical studies have shown a risk in which learners fail to monitor and control their own learning when learning with generative AI and instead offload these foundational SRL processes to AI (Darvishi et al., 2024; Fan et al., 2025; Stadler et al., 2024; Urban et al., 2025). This offloading takes the form of human-AI interactions that suppress learners' agency in learning (Brod,

2026): while learners make the decision to offload these metacognitive processes to generative AI, it is the AI that makes and enacts decisions about learning process.

In the context of supporting SRL, we view human-AI interactions as a form of shared regulation (Molenaar et al., 2023). Understanding human-AI interactions as reciprocal, where AI considers learners' internal conditions (e.g., prior domain knowledge, motivational beliefs) to tailor its invitations to self-regulate one's learning, and learners critically evaluate and act upon AI's invitations, enhances the learner's ability to purposefully self-regulate their own learning process. In practice, these invitations to self-regulate can include offloading metacognitive monitoring from learners to AI, after which controlling one's learning is onloaded from AI to learners. For example, learners may consult a learning-analytics driven dashboard indicating that they are off-track in achieving their learning goals, or receive a personalized scaffold that raises their metacognitive awareness about insufficient task understanding. Based on this increased metacognitive awareness, learners can revise their learning goals, adapt their cognitive strategies to achieve those goals, or review the task instructions, demonstrating metacognitive control. Thus, while AI augments learners in making decisions about their learning (i.e., metacognitive monitoring), learners themselves develop their capacity to enact those decisions (i.e., metacognitive control, Brod, 2026). This dynamic on- and offloading benefits both learners and AI when learners do not offload their regulatory practices to AI (Molenaar, 2022) but instead facilitate a co-evolving system in which AI and learners mutually adapt. This system supports the transition from adaptive learning environments to empowered learners whose SRL is augmented by AI.

This special issue uses the SMA grid as a starting point to explore the intersection of SRL, LA, and AI from the perspectives of measuring and supporting SRL, drawing both on reviewing the existing empirical evidence and providing new evidence-based insights on the topic. In this special issue, interdisciplinary research grounded in SRL theory highlights the dynamic interplay between CAMM processes, diverse data streams, and suitable analytical techniques, and provides suggestions on how these insights can be harnessed to provide SRL support that empowers learners. The current special issue includes three review studies and two empirical studies, focused on different aspects of SRL and how to apply the SMA grid to measure and support SRL processes.

Contributions of the Special Issue

The first review study by de Mooij et al. (2025) shows the mapping of studies in the SRL field onto the SMA grid (Molenaar et al., 2023). The review identified studies measuring CAMM processes during learning and examined the synthesis on which combinations of data streams and analytical approaches (unimodal, horizontal, vertical, and integrated) were employed across learning contexts and environments. The mapping on the SMA grid shows more integrated use of multimodal data and the dynamics between the CAMM processes, although motivational and affective processes remain underrepresented. Furthermore, they find that studies using integrated approaches mostly analyze different processes and data streams sequentially, and that

aligning the temporality of data streams to capture their dynamic relationship with CAMM processes is rarely done. The review also discusses the strengths of each analytical approach and AI-powered alignment methods to guide future SRL research.

The second contribution to the special issue is by Törmänen et al. (2025), who conducted a systematic review of 80 empirical studies to examine how situational academic motivation has been measured and conceptualized in recent research. Framed within five key motivation theories, including self-determination theory and expectancy-value theory, the review distinguished two major waves of research: one focusing on situation-specific measures of motivation ($n=62$), and a more recent wave capturing dynamic, moment-to-moment changes in motivation during learning ($n=18$). By mapping these studies, the authors expanded the motivational dimension of the SMA grid (de Mooij et al., 2025; Molenaar et al., 2023) by highlighting the role of context and time when studying motivation. The findings reveal that situational motivation is highly context-dependent, fluctuates within and between learners, and is shaped by complex, temporally embedded interactions among motivational constructs. However, most studies relied on self-report measures, and few employed time-sensitive or multimodal methods. The authors call for future research to explore a new process-oriented wave that integrates multimodal data of novel, unobtrusive online process measures.

Bühler et al. (2025) conducted a systematic literature review mapping mind wandering onto the SMA grid (de Mooij et al., 2025; Molenaar et al., 2023), highlighting its connections to SRL. They suggest that mind wandering can both prompt SRL (e.g., refocusing attention) and function as a regulatory strategy (e.g., managing frustration). In their review of 134 studies, they found that cognitive and metacognitive processes were frequently examined, whereas affective and motivational processes were underrepresented. Mind wandering was typically assessed via self-reports (e.g., probes), eye-tracking, or both. The authors applied and adapted SMA's data integration concepts: unimodal (one data stream), horizontal (mind wandering and CAMM in one stream), vertical (mind wandering across multiple streams), and integrated (mind wandering and CAMM across multiple streams). For instance, integrated studies might track task-unrelated thoughts alongside cognitive and metacognitive strategies by combining self-reports with eye-tracking. Though only about 25% of studies used AI, unobtrusive, machine-learning-based multimodal sensing provides a scalable way to continuously assess learners' attentive states. This assessment, together with real-time SRL measurements based on multimodal data, could be used in learning technologies that can diagnose whether mind wandering is used as a regulatory strategy or whether there is a need to invite SRL from learners.

Next, we introduce two empirical papers included in this special issue. First, Kubsch et al. (2025) took initial steps toward developing an AI-based early warning system that draws on students' cognitive artifacts, along with self-report data on affect and metacognition from inquiry-based science classrooms. They adopted an integrated approach, focusing on cognition, metacognition, and affect by combining log data from a technology-enhanced workbook (cognitive processes) with self-reports (metacognitive and affective processes). A notable strength of the study was its implementation in authentic classroom settings, embedded within an instructional unit spanning multiple lessons over 2–3 weeks. Interestingly, the authors found that the importance of different CAM dimensions (cognition, affect, and metacognition)

varied over time. While self-reports on affect and metacognition were more accurate in predicting learning productivity at the beginning of the instructional unit (based on students' scores across all unit tasks), the predictive accuracy of students' cognitive artifacts increased toward the end of the unit. By applying various machine learning methods, the authors found that boosting yielded the highest overall predictive accuracy. Boosting also showed the least bias regarding predictive accuracy by gender. However, when predictive accuracy was examined by parental education background, the decision tree model outperformed boosting. This study offers valuable insights into balancing predictive power and model explainability in early warning systems and other SRL-related support tools or interventions, highlighting the importance of fairness and transparency in practical AI applications in classrooms.

The final empirical contribution by Moreno et al. (2025) advances the field of SRL by employing supervised machine learning to predict medical trainees' physiological responses, measured via electrodermal activity (EDA), during distinct phases of high-fidelity simulation training. In this study, the authors again adopted an integrated approach, using video recordings, EDA, and self-reports to examine cognitive, metacognitive, and socio-emotional (affective) processes, to explore how SRL and socially shared regulation of learning (SSRL) behaviors among medical trainees during team simulations can predict their physiological responses. Sixty-two medical residents participated in simulations coded for behavioral and regulatory strategies, with EDA data collected via wearable sensors. Results showed that different patterns of SRL and SSRL behaviors were predictive of EDA at distinct stages of the simulation, highlighting how learners' regulatory needs shift over time. This work offers valuable insights into the integration of multimodal data and machine learning for capturing and supporting regulation in authentic, high stakes learning contexts.

Concluding Remarks

While an increasing number of SRL reviews are available (see, e.g., Saint et al., 2022; Sharma et al., 2024; Viberg et al., 2020), our special issue builds on systematic theoretical and empirical work on SRL operationalized through the SMA grid (e.g., Azevedo & Gašević, 2019; Järvelä & Hadwin, 2024). The contributions in this special issue demonstrate that the field is now mature enough to move beyond merely describing SRL processes, toward predicting and augmenting SRL across diverse contexts and domains, considering the full range of CAMM processes, including affective and motivational ones. While the application of AI in both the empirical studies and the studies included in the systematic reviews mostly deals with machine learning, we expect that the use of generative AI in detecting, diagnosing, and acting on SRL (Molenaar, 2022) will increase in the future.

In the proposal for this special issue, we posed a set of guiding questions for the contributing authors:

1. How can multimodal data be used to capture the dynamics of CAMM processes?
2. How can findings on CAMM processes, based on multimodal data analyzed with appropriate methods, be used to effectively support SRL?

3. How can AI methods be applied to analyze multimodal data to capture the dynamics of CAMM processes?
4. How can human–AI collaboration advance the support of SRL?

Table 1 provides a summary of how each paper in this special issue addresses these key questions. Collectively, these contributions set the stage for future research through

Table 1 Summaries of how the contributions address the key questions posed in the proposal for this special issue

	Contribution 1 de Mooij et al.	Contribution 2 Törmänen et al.	Contribution 3 Bühler et al.	Contribution 4 Kubsch et al.	Contribution 5 Moreno et al.
How can multimodal data be used to capture the dynamics of CAMM processes?	Researchers increasingly use integrated multimodal data, behavioral, physiological, and contextual, to capture how CAMM processes unfold and interact over time. Advanced methods like sequential analysis, process mining, and AI tools help model these dynamic patterns more accurately.	Measuring situational motivation through multimodal data in a “process-oriented wave”, as the paper explains, is rarely used, and there is a need to explore more multimodal online continuous measures apart from the learners’ subjective beliefs.	Relations between mind wandering, cognitive and metacognitive processes have mostly been studied through self-report and eye-tracking data.	Cost-effective data set including self-report data (informing affective and metacognitive processes) and artifact data from digital learning environments (informing cognitive processes) together predict learning productivity.	Varying SRL and SSRL codes and behaviors, such as posture and facial expressions derived from video and audio recordings, predicted emotional activation (measured with electrodermal activity, EDA) across different segments of the simulation.
How can AI methods be applied to analyse multimodal data to capture the dynamics of CAMM processes?	AI can analyze by aligning and modeling diverse data streams, like log files, eye tracking, and physiological signals, to detect patterns in CAMM processes over time. Techniques such as machine learning, natural language processing, and hidden Markov models enable more accurate, scalable, and dynamic analysis of SRL behaviors, but is not used so often yet.	Novel (AI) methods are particularly needed to capture the dynamics of motivational processes, which based on the findings are often overlooked but essential for understanding and supporting regulation in learning.	The use of multimodal data improves the accuracy of mind-wandering prediction and necessitates applying machine learning or other advanced analysis techniques.	By applying a range of machine learning techniques, a relatively simple and transparent method (i.e., boosting) was found to effectively capture the dynamics of CAM processes in predicting learning productivity.	Supervised machine learning (Lasso regression) was used, with the average EDA peaks per segment as the predicted variable and coded behaviors, regulation strategy counts, and self-reports as predictors across the three phases of the simulation task.

Table 1 (continued)

	Contribution 1 de Mooij et al.	Contribution 2 Törmänen et al.	Contribution 3 Bühler et al.	Contribution 4 Kubsch et al.	Contribution 5 Moreno et al.
How can the findings on CAMM processes based on multimodal data analysed with appropriate methods be used to support SRL effectively?	Findings from multimodal data on CAMM processes on the grid can guide support by identifying when learners face cognitive, affective, metacognitive, or motivational challenges. By integrating these insights into the detect-diagnose-act framework, learning systems can provide real-time, personalized interventions that augment self-regulated learning.	Understanding the fluctuations of motivation in real-time, is fundamental to provide personalized AI-support in the moment to student's motivational needs as part of SRL.	The use of multimodal data together with machine learning techniques can inform attention-aware learning technologies that provide real-time, personalized support for SRL.	The predictive power of the different CAM dimensions shifted over time that is essential for designing timely and targeted SRL support. The findings support the implementation of early-warning systems in real classrooms using easily accessible data and transparent machine learning techniques without requiring heavy technological investments.	Multimodal learning analytics, as applied in this study, could help tailor support so that, as students' cognitive load increases (indicated, for example, by emotional activation derived from physiological data) assistance in implementing appropriate regulatory strategies can be provided.
How can human-AI collaboration advance supporting SRL?	Studies can combine AI's efficiency in analyzing large-scale multimodal data with human expertise in interpreting learning behaviors. This synergy enables faster, more accurate detection of CAMM challenges and supports the development of adaptive, context-aware learning interventions.	Human-AI collaboration could be used to develop measures that capture momentary variations in learners' motivation. Such novel measures are necessary to provide timely support for students' motivational processes, which are underrepresented in current studies.	Human-AI collaboration approaches may help contextualize mind-wandering either as a purposeful regulatory response (e.g., managing frustration) or as an unconscious process that is an important distinction when supporting SRL.	Early warning systems can support SRL by enabling early identification of students at risk through accurate predictive models and by informing targeted, theory-aligned interventions via interpretable explanatory models. Predictive and explanatory models together allow teachers to understand not just who need support, but why. This facilitates timely and pedagogically meaningful SRL support while maintaining teachers' agency in human-AI interactions.	Human-AI interactions can be used to link learners' emotional activation with their regulatory strategies, allowing support for pro-social strategies such as facilitation and evaluation. These can then be reinforced through debriefings or training to help learners manage increased cognitive load and strengthen their SRL.

multidisciplinary collaboration, particularly in investigating human–AI interactions in SRL and hybrid intelligence solutions for augmenting SRL (Järvelä et al., 2025).

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